



The Sun: Our Extraordinary Ordinary Star

【【生词双高亮】BBC科普精选：太阳如何主宰地球命运】

Summary: The Sun, while an ordinary star in the vast galaxy, holds unparalleled significance for life on Earth. It not only enables life but deeply influences human biology, behavior, and modern technology. Our internal body clocks are synchronized by sunlight, and disruption to this rhythm can lead to serious health issues. The Sun also affects social behaviors like generosity and crime rates. Despite being a constant nuclear reactor, solar activities like storms pose potential risks to our technology-dependent civilization. Harnessing solar energy remains key to a sustainable future.

摘要：太阳虽是银河系中一颗普通恒星，却是地球生命存在的基石。它不仅赋予生命能量，更深刻影响着人类生理节律、行为模式乃至社会活动。阳光调控着人体生物钟，一旦失调可能引发肥胖、心脏病等健康危机。研究显示日照程度会影响消费意愿与犯罪率。这颗持续核聚变的恒星既蕴藏着满足全球能源需求的潜力，其爆发的太阳风暴也可能对现代科技文明造成毁灭性打击。开发利用太阳能已成为可持续发展的重要方向。





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▶ The Sun is just one of hundreds of billions of stars in our galaxy.

太阳只是我们#星系#中数千亿颗恒星之一。

▶ In a way, there's nothing very special about it.

从某种角度来说，它并没有什么特别之处。

▶ It's a fairly ordinary star, not particularly large or bright by stellar standards, but it's very special to us.

按#恒星#标准来看，它只是一颗相当普通的恒星，体积和亮度都不突出，但对我们却意义非凡。

▶ The Sun makes life on Earth possible.

太阳让地球上的生命成为可能。

▶ Cultures across the world and throughout the ages worship the Sun.

古今中外各种文化都崇拜太阳。

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▶ In some ways, modern life has cut us off from the Sun.

现代生活在某些方面让我们与太阳隔绝。

▶ In the developed world, we spend 90% of our time indoors.

在发达世界，我们90%的时间都在室内度过。

▶ And yet, the Sun is still a powerful force underpinning our lives.

然而太阳仍然是支撑我们生活的强大力量。

▶ The Sun affects our perception of beauty.

太阳影响着我们对美的#感知#。

▶ When people talk of the golden hour, that time just before the Sun sets, it's a real thing.

当人们谈论黄金时刻——即日落前的那一刻——这确实存在科学依据。

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▶ The Sun isn't actually yellow or orange or even red.

太阳实际上并非黄色、橙色甚至红色。

▶ It's all the colours mixed together.

它是所有颜色的混合体。

▶ When the Sun rises or sets in the sky, the shortwave colours, green, blue and violet, are scattered, leaving only the yellow and the red part of the spectrum, giving that amazing glow.

当太阳升起或落下时，绿色、蓝色和紫色等短波色彩被散射，只留下光谱中的黄色和红色部分，形成那种绚丽的霞光。

▶ When the Sun is high in the sky, its blue waves bounce around, which is why the sky looks blue.

当太阳高悬时，其蓝色光波四处#弹射#，这就是天空呈现蓝色的原因。

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▶ And when you see a rainbow, that's the light from the Sun separated into all its magnificent colours.

当你看到#彩虹#时，那是太阳光分离成所有#绚烂#色彩的现象。

▶ Although we talk of the Sun rising in the East and setting in the West, that's not quite true.

虽然我们常说太阳东升西落，但这并不完全准确。

▶ It just seems that way to us.

这只是我们视角中的表象。

▶ The Sun stays in the same place.

太阳始终处于原位。

▶ It's the Earth that rotates on its axis.

是地球在绕着它的#轴#自转。

▶ This movement of the Sun is deeply embedded in our biology.

这种太阳运动规律深深烙印在我们的生物机制中。

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▶ When sunlight enters the eye, it sends signals to a master clock in our brain called the Supra-Chiasmatic Nucleus.

当#太阳光#进入眼睛时，会向大脑中的主生物钟——视交叉上#核#发送信号。

▶ This internal clock regulates everything, from when we sleep to how we digest a meal.

这个内在时钟调节着从睡眠时间到食物#消化#方式的一切生理活动。

🎵 Messing with this finely tuned machine, when we work night shifts or fly across the world, can make us feel pretty rough.

当我们上夜班或环球飞行时，打乱这个精密机制会让人倍感不适。

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🎵 Even the blueish light from a mobile phone late at night is enough to disrupt and confuse our internal body clock.

深夜手机发出的蓝光也足以#扰乱#我们的内在生物钟。

🎵 Being out of step with the Sun affects our mood and our ability to think clearly, and there's evidence that this kind of disruption can lead to higher rates of obesity, heart disease, diabetes and even cancer.

与太阳不同步会影响情绪和思维清晰度，有证据表明这种紊乱可能导致肥胖症、心脏病、糖尿病甚至癌症的发病率升高。

🎵 We're living out of step with the Sun, and some scientists say this could be causing a public health crisis.

我们正过着与太阳节律脱节的生活，有科学家称这可能引发公共健康危机。

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🎵 When the Sun's shining, studies have shown that we tip more generously and are more likely to splash out on luxury goods.

研究表明，阳光明媚时人们给小费更慷慨，更愿意#挥霍#购买#奢侈品#。

🎵 And crime may go down too.

犯罪率也可能下降。

🎵 One study in the US found that when the clocks go forward for daylight savings time, the number of robberies, rapes and murders went down around 50%.

美国一项研究发现，夏令时时钟拨快后，抢劫、强奸和谋杀案发率下降约50%。

🎵 The Sun is a hell of a force to be reckoned with.

太阳确实是一股不容小觑的强大力量。

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▶ Every hour, enough sunlight falls on the earth to power the world for a year.

每小时照射到地球的#阳光#足以满足全球一整年的能源需求。

▶ A year.

整整一年。

▶ Deep in its core, at around 15 million degrees Celsius, the Sun is constantly fusing hydrogen together to make helium.

在1500万摄氏度的核心深处，太阳持续将#氢#聚变成#氦#。

▶ These nuclear reactions release vast amounts of energy in the form of sunlight.

这些核反应以#太阳光#的形式释放巨大能量。

▶ If only we could work out how to harness this.

若能掌握#利用#这种能量的方法该多好。

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▶ But we are getting there.

但我们正在取得进展。

▶ It's estimated that by 2022, 30% of our energy could come from renewables, with solar growing faster than anything else.

预计到2022年，30%的能源将来自可再生能源，其中太阳能增速最快。

▶ The Sun is vast, dynamic and sometimes violent.

太阳浩瀚无边、#充满活力#，有时甚至很狂暴。

▶ Viewed up close, the Sun's surface looks like a raging sea of fire, with huge eruptions rising hundreds of thousands of kilometers into space.

近距离观察，太阳表面如同翻腾的火海，巨大的日珥喷发高达数十万公里。

▶ There are even earthquakes on the Sun, called Sunquakes.

太阳上甚至还有被称为"日震"的地震现象。

▶ And what happens on the Sun can impact us.

太阳的活动会影响我们。

▶ The Sun sporadically blasts huge clouds of charged particles towards the Earth, potentially wreaking havoc on the technology that we all rely on.

太阳会间歇性向地球喷射大量带电粒子云，可能对我们依赖的技术造成#严重破坏#。

▶ But 97% of people have never heard of solar storms.

但97%的人从未听说过太阳风暴。

▶ In 1859, a gigantic solar storm hit the Earth, creating auroras that covered the entire planet, knocking out the telegraph network and sending sparks flying from electrical equipment.

1859年，一场巨型太阳风暴袭击地球，产生的极光笼罩全球，导致电报网络瘫痪，电气设备火花四溅。

▶ The solar storm of 1859 is still the largest on record.

1859年太阳风暴仍是有记录以来最强烈的一次。

▶ But if a similar storm were to hit the Earth today, the consequences for our way of life could be devastating.

若类似风暴今日来袭，对我们生活方式的冲击将是毁灭性的。

▶ Knocking out electricity grids, satellite navigation and communications, for days, weeks or even months.

可能导致电网、#卫星##导航#和通信系统瘫痪数日、数周甚至数月。

▶ No wonder people used to worship the Sun.

难怪古人曾崇拜太阳。

